

No MUB quadruple in dimension six from Butson Hadamard matrices of root order 24

A certified finite reduction (draft for verification)

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Abstract

We prove that no four mutually unbiased bases of $\mathbb{C}^6$ have the form $\{I, H_1/\sqrt{6}, H_2/\sqrt{6}, H_3/\sqrt{6}\}$ with every entry of H_1, H_2, H_3 a 24th root of unity; equivalently, no MU quadruple in dimension six can be built from Butson Hadamard matrices in $\text{BH}(6, 24)$. The proof is a reduction of the statement to a finite assertion (Proposition 6), checked by an exhaustive exact computation in $\mathbb{Z}[\zeta_{24}]$ carried out by three independently written programs. This confirms, in certified exact arithmetic, the numerical search of Bengtsson et al. [3] at 24th roots, and extends the published theorem record, which stops at $\text{BH}(6, 12)$ [6, Thm. 7.15]. As a by-product we obtain the first classification of $\text{BH}(6, 24)$: it has exactly 25 monomial-equivalence classes (14 up to transpose). This note contains the complete reduction argument and the algorithm contract needed to re-implement the computation independently; the accompanying repository contains code, deterministic logs, and crosswalks to independently published data.

1 Introduction

Two orthonormal bases $\{e_i\}, \{f_j\}$ of $\mathbb{C}^d$ are *mutually unbiased* (MU) if $|\langle e_i | f_j \rangle|^2 = 1/d$ for all i, j . In every prime-power dimension exactly $d+1$ MU bases exist; dimension six is the first open case, and Zauner [1] conjectured that no more than three MU bases exist in $\mathbb{C}^6$. Writing bases in Hadamard form, a hypothetical fourth basis yields an *MU quadruple* $\{I, H_1/\sqrt{6}, H_2/\sqrt{6}, H_3/\sqrt{6}\}$ where each H_k is a complex Hadamard matrix of order six and the three are pairwise mutually unbiased. Progress on the conjecture typically takes the form of excluding strata of the complex Hadamard landscape from participating in a quadruple.

A *Butson Hadamard matrix* $H \in \text{BH}(d, N)$ is a $d \times d$ matrix with all entries N th roots of unity and $HH^* = dI$ [2]. These discrete strata admit finite exact exhaustion. Bengtsson et al. [3] searched them numerically in 2007: they enumerated the order-six Hadamard matrices with 12th- and 24th-root entries, the vectors unbiased to each, the third bases these admit, and whether any two third bases yield four MU bases, finding only triplets. The survey [6] records the resulting theorem only at root order 12 (its Theorem 7.15). The present note upgrades the 24th-root exclusion to a certified theorem.

Theorem 1. *There is no MU quadruple $\{I, H_1/\sqrt{6}, H_2/\sqrt{6}, H_3/\sqrt{6}\}$ in $\mathbb{C}^6$ with $H_1, H_2, H_3 \in \text{BH}(6, 24)$.*

Since $\zeta_m = \zeta_{24}^{24/m}$ embeds the m th roots into the 24th roots for every $m \mid 24$, the theorem covers all entries drawn from roots of unity of order dividing 24, in any mixture (Corollary 7), and a small additional computation extends the exclusion to every single root order $N \leq 12$ (Corollary 8). The theorem says nothing about other root orders or about the continuous Hadamard families, and so does not resolve Zauner's conjecture. The stratum is not vacuous: it contains genuine MU triplets, and the computation in fact enumerates all 1,920 of them and shows none extends.

2 Notation and elementary reductions

Fix $N = 24$, $\omega = e^{2\pi i/24}$, and $d = 6$. We work with *exponent vectors* $a \in (\mathbb{Z}/24)^6$, representing the unimodular vector $\omega^a = (\omega^{a_0}, \dots, \omega^{a_5}) \in \mathbb{C}^6$. A vector (or matrix row or column) is *dephased* if $a_0 = 0$. For $a \in (\mathbb{Z}/24)^6$ put

$$S(a) = \sum_{j=0}^5 \omega^{a_j} \in \mathbb{Z}[\omega].$$

We say a satisfies the *Z-condition* if $S(a) = 0$ and the *M-condition* if $|S(a)|^2 = 6$. Both conditions are invariant under adding a constant to all coordinates of a , since this multiplies $S(a)$ by a root of unity. For $u, w \in (\mathbb{Z}/24)^6$,

$$\langle \omega^u | \omega^w \rangle = \sum_j \overline{\omega^{u_j}} \omega^{w_j} = S(w - u),$$

so two unimodular root-valued vectors are *orthogonal* iff their exponent difference satisfies the Z-condition, and *unbiased* (for bases scaled by $1/\sqrt{6}$) iff it satisfies the M-condition. All tests are exact: $\mathbb{Z}[\omega] \cong \mathbb{Z}[x]/\Phi_{24}(x)$ with $\Phi_{24}(x) = x^8 - x^4 + 1$, so $S(a)$ and $|S(a)|^2 = S(a)\overline{S(a)}$ are integer vectors of length 8, and the conditions are integer identities. Define the dephased ray classes

$$Z = \{a \in \{0\} \times (\mathbb{Z}/24)^5 : S(a) = 0\}, \quad M = \{a \in \{0\} \times (\mathbb{Z}/24)^5 : |S(a)|^2 = 6\}.$$

Note that any unimodular vector is automatically unbiased to every standard basis vector, so unbiasedness to I is free, and that a 6×6 matrix with unimodular entries has orthogonal columns iff it has orthogonal rows (both say $\frac{1}{\sqrt{6}}H$ is unitary).

Lemma 2 (Normal form). *If an MU quadruple as in Theorem 1 exists, then one exists in which additionally H_1 is dephased (first row and first column all ones) and every column of H_2 and H_3 is dephased.*

Proof. Let D_L be the diagonal unitary whose j th entry is the conjugate of the j th entry of the first column of H_1 ; its entries are 24th roots. Replacing each basis B of the quadruple by $D_L B$ preserves all MU relations (a common unitary) and maps the standard basis to itself up to phases; it multiplies each H_k on the left by D_L , keeping all entries in μ_{24} and making the first column of H_1 all ones. Next, multiplying any H_k on the *right* by a diagonal matrix of 24th roots rephases its columns, which changes neither the basis it represents nor membership in $\text{BH}(6, 24)$: use this to make the first row of H_1 all ones, and to dephase every column of H_2 and H_3 . $\square$

Lemma 3 (First basis). *The map sending a dephased $H \in \text{BH}(6, 24)$ to the set of its five non-initial rows is a bijection between dephased $\text{BH}(6, 24)$ matrices with sorted rows and 5-element subsets $R \subseteq Z$ whose pairwise differences satisfy the Z -condition. Permuting the non-initial rows of H_1 preserves the quadruple property, so it suffices to enumerate the subsets R .*

Proof. A dephased H has first row ω^0 (all ones) and every row r with $r_0 = 0$. Orthogonality of row r against the all-ones row is $S(r) = 0$, i.e., $r \in Z$; orthogonality of rows r, r' is the Z -condition on $r' - r$, which is again dephased. Conversely such a subset, together with the all-ones row, has pairwise orthogonal unimodular rows, hence lies in $\text{BH}(6, 24)$. Row permutations fixing the first row amount to left multiplication by a permutation matrix, which fixes the standard basis and every MU relation. $\square$

Lemma 4 (Second and third bases). *Fix a dephased H_1 with columns $h^{(0)} = 0, h^{(1)}, \dots, h^{(5)}$ (exponent vectors), and define the completion rays*

$$U(H_1) = \{c \in M : c - h^{(i)} \text{ satisfies the } M\text{-condition for } i = 1, \dots, 5\}.$$

Then the bases unbiased to both I and $H_1/\sqrt{6}$ that are representable with columns in $\text{BH}(6, 24)$ form (after column dephasing and up to column order) exactly the 6-element subsets $C \subseteq U(H_1)$ whose pairwise differences satisfy the Z -condition.

Proof. By Lemma 2 each column c of such a basis matrix may be taken dephased with entries in μ_{24} . Unbiasedness against the column $h^{(0)} = 0$ of H_1 is the M -condition on $c - 0$, i.e., $c \in M$; against the remaining columns it is the M -condition on $c - h^{(i)}$ (the M -condition is phase-invariant, so the non-dephased difference is immaterial). Hence each column lies in $U(H_1)$. Orthogonality of two columns is the Z -condition on their difference. Distinct columns have distinct dephased exponent vectors (equal ones are not orthogonal), so the six columns form a 6-subset; column order is immaterial for the basis. The converse direction is immediate. $\square$

Lemma 5 (Pair test). *With H_1 fixed as above, two bases given by 6-cliques $C_2 \neq C_3 \subseteq U(H_1)$ are mutually unbiased iff every difference $c' - c$ with $c \in C_2, c' \in C_3$ satisfies the M -condition.*

Proof. Immediate from the correspondence between unbiasedness and the M -condition on exponent differences. (Note $C_2 = C_3$ can never yield an MU pair: a basis is not unbiased to itself, and indeed $S(0) = 6$ fails the M -condition.) $\square$

Proposition 6 (Finite reduction). *Theorem 1 holds if and only if for every 5-subset R as in Lemma 3 (equivalently, every dephased H_1), no two distinct 6-cliques $C_2, C_3 \subseteq U(H_1)$ have all 36 cross-differences satisfying the M -condition.*

Proof. Lemmas 2–5 map any MU quadruple in the stratum to such a configuration and conversely. $\square$

3 The computation

The finite assertion of Proposition 6 is checked by the following five-stage algorithm; the stage-by-stage counts are part of the claim and serve as checkpoints for independent re-implementations.

1. **Ray classification.** For each of the $24^5 = 7,962,624$ dephased exponent vectors a , decide the Z - and M -conditions exactly in $\mathbb{Z}[x]/\Phi_{24}(x)$. Result: $|Z| = 8,350$ and $|M| = 19,320$.

2. **Hadamard enumeration.** Enumerate all 5-subsets of Z with pairwise differences in the Z -condition. Result: exactly 7,584 dephased $\text{BH}(6, 24)$ row sets.
3. **Completion rays.** For each row set, compute $U(H_1)$ by the column tests of Lemma 4. Census over the 7,584 matrices, written $|U|/\#\text{bases}$: $0/0 : 6,852$; $6/1 : 240$; $12/4 : 60$; $24/4 : 360$; $40/0 : 72$.
4. **Completion bases.** Enumerate the 6-cliques of Lemma 4. Result: 1,920 bases, spread over 660 matrices ($240 \times 1 + 60 \times 4 + 360 \times 4$), giving $420 \times \binom{4}{2} = 2,520$ candidate pairs.
5. **Pair refutation.** For each of the 2,520 pairs, exhibit a cross pair (c, c') whose difference fails the M -condition. All 2,520 pairs are refuted; no MU quadruple exists in the stratum.

Three independently written programs (available in the accompanying repository, with deterministic logs and SHA-256 manifests) implement this contract: the original search, an adversarial referee harness that re-derives the Φ_{24} reduction table from a recurrence and verifies all bases and refutations in a second exact representation, and a third, code-independent enumerator. All agree on every count above. For re-implementers preferring floating point: over all 24^5 rays the smallest margin of a non-zero-sum ray from 0 is 0.035 and of a non-unbiased ray from unbiasedness is 1.2×10^{-3} , with no ray in an ambiguous band at double precision; the certified programs use exact arithmetic throughout.

4 Corollaries

Corollary 7. *No MU quadruple $\{I, H_1/\sqrt{6}, H_2/\sqrt{6}, H_3/\sqrt{6}\}$ exists in which every entry of H_1, H_2, H_3 is a root of unity of order dividing 24 (the orders may vary from entry to entry).*

Proof. All entries then lie in μ_{24} , so each $H_k \in \text{BH}(6, 24)$ and Theorem 1 applies. $\square$

Corollary 8. *For every single root order $N \leq 12$, no MU quadruple exists with all entries of H_1, H_2, H_3 being N th roots of unity.*

Proof. For $N \in \{1, 2, 3, 4, 6, 8, 12\}$ this is Corollary 7. For $N \in \{5, 7, 11\}$ a vanishing sum of six N th roots of unity would have length divisible by the prime N (Lam–Leung [4]), so no row can be orthogonal to the all-ones row and $\text{BH}(6, N) = \emptyset$. For $N \in \{9, 10\}$, direct computation (seconds; included in the repository) shows the analogue of M is empty: no dephased N th-root vector is unbiased to the all-ones vector. Dephasing as in Lemma 2 stays inside μ_N , so no basis can be unbiased to both I and a dephased H_1 , and not even an MU triple exists in these strata. $\square$

5 External anchors and the $\text{BH}(6, 24)$ classification

Every stage of the computation reproduces independently published data.

Hadamard enumeration. Closing the 7,584 dephased row sets (and the analogous enumerations at smaller orders) under the full residual group of monomial equivalence yields 1, 1, 4, 3, 11 equivalence classes at $N = 3, 4, 6, 8, 12$, matching the published classification of Lampio, Östergård and Szöllősi [5] at every order they tabulate, and **25** classes at $N = 24$ (14 up to transpose), for which no published value exists. Of the 25 classes, exactly 4, 3, and 11

contain representatives of $\text{BH}(6, 6)$, $\text{BH}(6, 8)$, and $\text{BH}(6, 12)$ respectively, and 12 are new at root order 24. Representatives with their completion data accompany this note.

Completion census. Exactly five of the 25 classes admit at least one completion basis, with per-matrix base counts 4, 1, 1, 4, 4; they are identified as the classes of $F(0, 0)$, $F(1/6, 0)$, $F^T(1/6, 0)$, $F(1/6, 1/12)$, and $D(1/8)$, matching Fig. 2 of [3] class for class (one further class has 40 completion rays but no basis, consistently omitted there). The agreement of the exact census with the independent 2007 numerical census is a strong mutual consistency check.

Theorem record. Restricting the pipeline to $N = 12$ reproduces the setting of [6, Thm. 7.15]: 2,184 dephased matrices, completion census $0/0 : 1,884$, $6/1 : 240$, $12/4 : 60$, 480 completion bases, and no unbiased pair.

6 Verification

The reduction argument of this note is elementary and short; we invite readers to check it directly. The finite assertion can be verified either by running the accompanying programs (minutes; deterministic logs and hashes provided) or, preferably, by independent re-implementation from the contract of Section 3, using the stage counts and the anchors of Section 5 as checkpoints. The computations were authored and cross-verified by three independent AI systems under a human-directed protocol described in the repository; confirmations and refutations are equally welcome and will be credited.

References

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